Data-Driven Optimization for Inventory, Credit Risk, and Supply Chain Management in New-A-One Bangle Store

A Proposal for the BDM Capstone Project

Submitted by

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Declaration Statement

I am working on a Project Title **"Data-Driven Optimization for Inventory, Credit Risk, and Supply Chain Management of New-A-One Bangle Store.**

". I extend my appreciation to New-A-One Bangles Store, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analysed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate: 

Name: Aqdas Khan

Date: 3rd February 2025

1. Executive Summary and Title

This project focuses on optimizing the operations of a wholesale bangles business named "New-A-One Bangle Store" situated in Sheikh Latif, Firozabad, Uttar Pradesh, India that sources bangles from manufacturers and supplies them to retailers and bulk buyers.

Despite the growing market demand, the business faces challenges related to inventory mismanagement, credit risks, and supply chain inefficiencies. Overstocking of slow-moving designs and stock shortages of high-demand products lead to financial losses. Additionally, delayed customer payments create cash flow constraints, while inconsistent product quality and supplier delays disrupt business operations.

To address these issues, I will perform in-depth data analysis that will leverage data-driven strategies to enhance inventory management, credit risk assessment, and supply chain optimization. The proposed solution involves implementing predictive analytics for demand forecasting, a credit scoring system to evaluate payment behaviour, and a supplier performance tracking framework. These measures will help improve stock availability, minimize financial risks, and enhance operational efficiency.

By implementing this data management system, the business is expected to achieve reduction in overstocking and wastage, faster payment cycles through structured credit management and automated reminders, enhanced supply chain efficiency with better supplier coordination and quality control.

2. Organization Background

New-A-One Bangle Store was established in 2018 in Sheikh Latif, Firozabad by Mr. Izhar Khan . The Business is operational for more than 7 years now. It operates as a wholesale distributor, procuring bangles from manufacturers and supplying them to retailers and bulk buyers. The business has built a strong reputation for offering a wide variety of high-quality bangles to meet diverse market demands. The model of the business is B2B.

The business is managed by a team of three staff members excluding the owner, Mr. Izhar Khan who oversee daily operations, including inventory management, order processing and other related tasks. The business has a large variety of traditional glass Bangles of many different colours. Firozabad, known as the bangle capital of India, provides an ideal location for sourcing high-quality products. the business focuses on maintaining product quality,

ensuring timely deliveries, and offering competitive pricing to its customers. The data is provided by Mr. Izhar Khan who is also my father that helped me getting my project started.

3. Problem Statement (Listed as objectives)

The business faces three major challenges that impact growth and profitability:

1. **Inventory Optimization Problem:**
    - Overstocking of slow-moving designs leads to capital blockage and increased storage costs.
    - Stockouts of high-demand bangles result in lost sales opportunities.
    - Lack of data-driven demand forecasting results in inefficient purchasing decisions.
2. **Credit and Payment Risk Management Problem:**
    - Many retailers and bulk buyers purchase on credit, leading to cash flow constraints.
    - Delayed or non-payments result in financial instability and potential bad debts.
    - Absence of a structured credit assessment system increases financial risk.
3. **Supply Chain Management Problem:**
    - Inconsistent product quality from different manufacturers leads to customer dissatisfaction.
    - Delays in supplier deliveries disrupt order fulfilment timeline.

4. Background of Problem

The New-A-One Bangle Store operates in a competitive market with challenges in inventory management, credit risk, and supply chain efficiency. The lack of a data-driven approach leads to financial and operational inefficiencies. Overstocking results in blocked capital, while understocking causes missed sales opportunities. Additionally, credit sales without structured risk assessment leads to cash flow issues. Supply chain inefficiencies, including inconsistent supplier performance and delays, further impact business operations.

Internal Factors

- Poor inventory tracking leads to over-purchasing of slow-moving designs and stockouts of high-demand products.

- Delayed payments from customers cause cash flow constraints, making it difficult to maintain steady operations.
- Lack of a credit evaluation system increases the risk of financial losses due to unpaid invoices.
- Inconsistent order fulfilment affects customer trust and retention.

External Factors

- Market competition forces the business to maintain low profit margins.
- Supplier delays impact the ability to meet customer demand on time.
- Economic fluctuations and changing customer preferences make demand prediction difficult.

5. Problem Solving Approach

Methodology used:

There might be many different methods that can be used to solve the business problems**.**

- **Inventory Analysis**: A study of past sales data that can be helpful to identify fast moving stock and slow moving stock. This can help in overcoming the problem of understocks and overstocks and maintain optimal inventory levels.
- **Credit Risk Assessments** :Developing a system to track costumer payment and assess credit risk that can help to overcome the problem of delayed payments and financial loses.
- **Supply Chain Improvement**: Analyzing suppliers performance and delivery timelines to ensure good quality and timely restocks.

Intended Data Collection:

The business has agreed to provide me their sales data, costumer purchase history data and suppliers data but the data is not digitized so I have to manually enter the records in excel and do the further analysis. The timeframe for the data spans from June 2024 to December 2024.

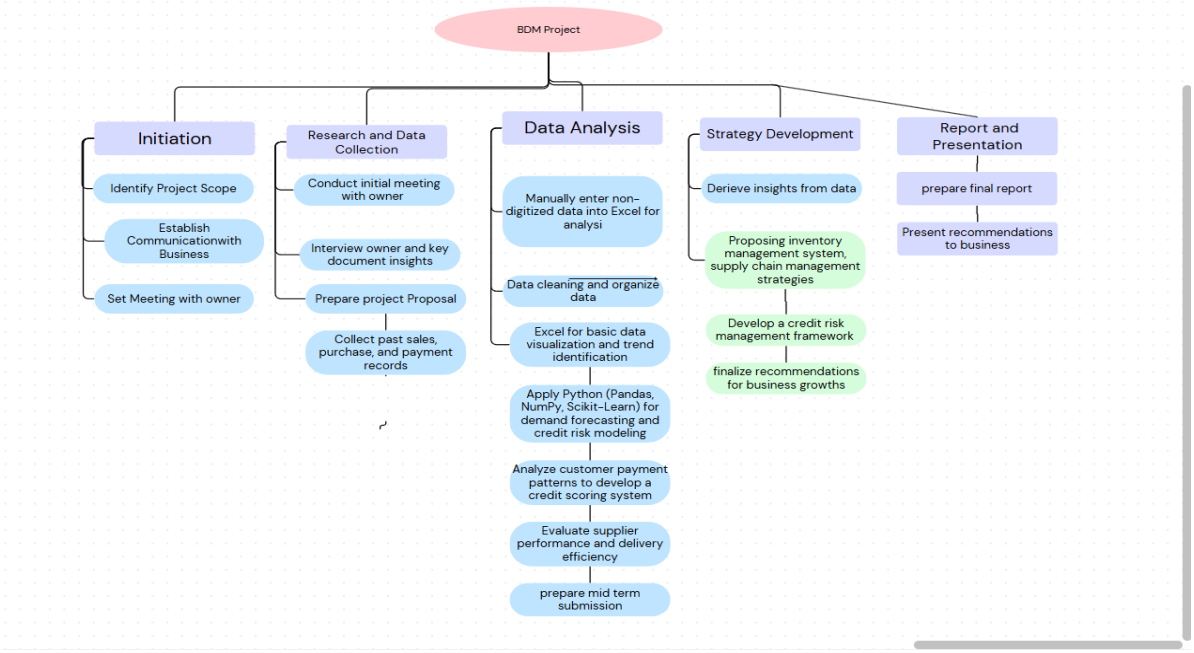
- **Sales data:** Collecting the past months sales data to identify the trends and understanding the demand pattern.
- **Costumer Payment Records**: Analyzing costumers payment records to identify reliable costumers.

- **Suppliers Data**: Collecting data on suppliers reliability and delivery times to optimize inventory.

Tools for analysis:

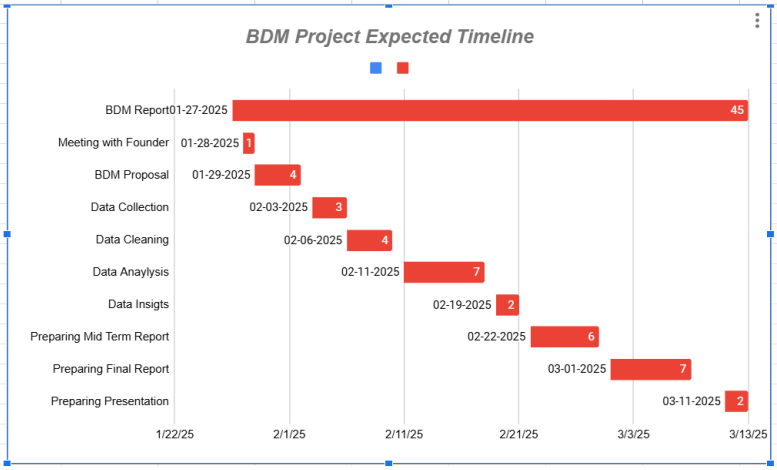
- **MS Excel:** Excel will be used for initial data entry and storage since the provided business data is not digitized. Its user-friendly interface allows for easy manual data entry, basic cleaning, and descriptive statistical analysis. Moreover, Excel's built-in pivot tables and charting tools will help visualize trends in inventory, sales, and credit risk.
- **Python:** Python with libraries like Panda, Numpy will be used for in depth data analysis. Credit risk assessment will be implemented using another yet useful python library knowns as Scikit-Learn.
- **Tableau**: Tableau will be used to create informative yet attractive dashboards for real-time data visualization. It will help monitor sales performance, inventory levels, and customer credit risk, providing actionable insights for business decisions.

6. Expected Timeline



Work Breakdown Structure

Task	Start Date	End Date	Duration
BDM Report	01-27-2025	03-13-2025	45
Meeting with Founder	01-28-2025	01-28-2025	1
BDM Proposal	01-29-2025	02-02-2025	4
Data Collection	02-03-2025	02-05-2025	3
Data Cleaning	02-06-2025	02-10-2025	4
Data Anaylysis	02-11-2025	02-18-2025	7
Data Insigts	02-19-2025	02-21-2025	2
Preparing Mid Term Report	02-22-2025	02-28-2025	6
Preparing Final Report	03-01-2025	03-08-2025	7
Preparing Presentation	03-11-2025	03-13-2025	2



Gantt Chart

7. Expected Outcome

i. **Better Inventory Management**: The analysis will provide clear understanding of sales trends , the business can reduce overstocking and minimize wastage. Stock Planning will help ensure that fast-moving items are always available while reducing capital tied up in slow-moving stock.

ii. **Better Credit Risk Management:** The introduction of a credit scoring system will help assess customer payment behaviour and minimize bad debt risks**.**

iii. **Better Supply Chain Efficiency:** By analyzing suppliers performance and delivery timelines, the business can identify with the most reliable manufacturers. This can reduce delays and ensure good product quality, ensuring smoother operation and high costumers satisfaction.

iv. **Increased Business Growth & Stability:** With better inventory management, credit risk management and supply chain management, business will experience high profitability and smother operational efficiency